AI or Not?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

TEACHER GUIDE · 1 OF 9

AI or Not?

> Prepare the resources, check readiness and choose a realistic teaching schedule before starting the explanation.

Teach the learning cycle

Primary target: Distinguish an observed result from an explanation of how it was produced

Session	Sequence and minutes	Resources
Session 1 · Understand · 45 min	0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.	Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.
Session 2 · Practise · 45 min	0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.	Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.
Session 3 · Investigate and transfer · 45 min	0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.	Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

A classroom clock speaks the time. Write one thing you observed and one thing you cannot determine just from hearing it.

You observed spoken time. Speech alone does not tell you whether the message was selected by a timetable, a learned system or a person.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

> A bus displays a delay **prediction** learned from past journeys. A **written rule** turns the display red above 10 minutes. A driver can override the message. Identify all three **mechanism**s.
>
> Would seeing a red display prove that learning occurred?

Concept to explain

Look for **evidence** of how the system reasons or learns, not just its label or screen.

Term	Meaning
AI system	a system using learned patterns or symbolic reasoning
fixed rule	an instruction written in advance
human-controlled	a person makes the important choice
evidence	a clue that supports or changes a claim

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Compact overview · 80-minute option

> This is an alternative compact allocation. The main starting plan above uses three 45-minute sessions; allow more time for the full investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Avoid making AI a yes-or-no property of a whole product before defining the component and decision. Modern **learned model**s and symbolic reasoning both belong in the broader AI discussion.

The learning goal here is **evidence**-based **mechanism** classification, not memorising a list of smart objects. Do not infer **training** from responsiveness, language, movement or one correct **output**.

Readiness answer

You observed spoken time. Speech alone does not tell you whether the message was selected by a timetable, a learned system or a person.

Resources to prepare

The bus diagram and photo-library case from the student chapter

Blank paper and three colours of pencil

A copy of workbook tasks W1 to W5; no online accounts or real student data

End goal for the pictorial case

By the end, I can separate a learned **prediction**, a **written rule** and a **human decision**.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

TEACHER GUIDE · 2 OF 9

Explain the mechanism

> Teach the information path, distinguish the three mechanisms, and model how evidence supports a limited conclusion.

The organising idea

Follow the information instead of imagining magic inside the machine.

1. Input — information entering

INPUT a command, signal, photo, sound, or number

2. Process — the deciding step

PROCESS a human choice, fixed instruction, or learned pattern

3. Output — the result

OUTPUT movement, calculation, action, or **prediction**

The principle to teach

CORE IDEA Look for **evidence** of how the system reasons or learns, not just its label or screen.

Mechanism, not appearance

UNDER THE HOOD A system should be classified by its decision **mechanism**. **Human control**, **fixed automation**, and model-based **prediction** can produce similar visible behavior, so appearance and marketing words are weak **evidence**.

Draw a fresh example with all three stages:

Connect a claim to its evidence

Every step connects a claim to observable **evidence**.

The case to investigate

CASE A remote-control car, calculator, motion light, smart speaker, photo sorter, and robot vacuum may all look clever, but they do not make decisions in the same way.

Step 1

STEP 1 — OBSERVE a command, signal, photo, sound, or number

Step 2

STEP 2 — TRACE a human choice, fixed instruction, or learned pattern

Step 3

STEP 3 — CONCLUDE movement, calculation, action, or **prediction**

Model answer

MODEL REASONING Follow the information and identify who or what makes the important choice. A robot body, screen, voice, or the word smart cannot prove AI.

If two **mechanism**s fit the **evidence**, say that **more information** is needed.

A claim to avoid

DO NOT OVERCLAIM Do not call a system AI because it looks, moves, or talks like a person.

What new **evidence** could change this answer?

Teacher precision note

This course concentrates on modern machine learning. AI also includes symbolic and rule-based approaches. A simple automatic switch or robot body is not, by itself, **evidence** of AI.

Discuss how the system reasons or learns, rather than judging appearance.

Concrete teaching case

Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

Classify all six using **mechanism evidence**. Which two need **more information**?

Exact explanation to model

Reasoning for the six devices

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

Explain the added worked example

Worked example · two explanations of the same display

The school bus display says DELAY **12 MINUTES** in red. Leila calls it AI because it predicted the delay. Arun calls it a simple rule because the number turned red.

Both have noticed a real part of the product, but neither has explained the whole system. To investigate, ask which decision you are talking about and what **evidence** reveals how that decision was made.

Separate the result from the mechanism

An **output** is what a system produces. A **mechanism** is how it produces that **output**. Seeing a red number establishes its colour; it does not reveal where the number came from.

Similar **output**s can come from different **mechanism**s, and one product can contain several **mechanism**s.

The supplied engineering evidence

For this lesson we have an engineering note: the delay estimator was fitted using records of earlier bus journeys; a person wrote the colour instruction; and the driver can replace the message.

These are supplied facts for our classroom case. In a real investigation, we would need trustworthy documentation, a demonstration of the **process** or another source of **mechanism evidence**.

Explain the three decisions separately

In the first row, **fitting** means adjusting a model using examples. Applying the fitted model to a new journey is a different step. The model can produce an incorrect estimate even if it was trained carefully.

In the second row, a written threshold compares the estimate with 10. It can follow its instructions perfectly while using a bad estimate. In the third row, a person makes the message decision.

> **Precision · rules can also be part of AI**
>
> AI is a broader field than learning from examples. It also includes approaches that represent knowledge and reason with rules. Therefore, do not use the shortcut that every rule is non-AI.
>
> In this case, a simple colour switch is ordinary automation; identifying that switch does not classify every other part of the bus system.

Key terms

Term	Meaning
Mechanism	The process that produces a result
Written rule	An instruction specified by a person before the system applies it
Learned model	A system whose adjustable values were fitted using examples
Evidence	Information that supports or challenges a particular claim
Hybrid system	A product that combines more than one kind of process

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Think aloud · observation → evidence → conclusion

Question: Does the red display prove that a **learned model** was used? First separate the claim from the **observation**. The **observation** is that the displayed number is red. The claim concerns the **process** that produced the number.

Next find relevant **evidence**. The engineering note says earlier journeys were used to fit the delay estimator. That note supports a learned-model classification for the estimator. The red colour alone does not.

Claim	Relevant evidence	Conclusion
The estimator uses a learned model	The note describes fitting on past journeys	Supported for the estimator
The red colour proves learning	Only the displayed colour is observed	Not supported by colour alone
The driver override is a model prediction	A driver selects the replacement message	Contradicted in this case

Model the limit of the conclusion

A strong answer states its limit: I can identify these three **mechanism**s from the supplied note, but I cannot infer the model's accuracy from one estimate.

Even if the bus arrives exactly **12 minutes** late, one success does not establish that the estimator will always work.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Teach with the picture and the explanation together

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: AI or Not?

> A fictional school umbrella display shows TAKE ONE before lunch. What produced that advice?

The facts we will use

A weather model was fitted on earlier weather records; its forecast is **70%** rain.

A **written rule** shows TAKE ONE at rain forecasts of **60%** or more. The teacher decides whether to distribute umbrellas.

AI ACADEMY / AI-1/1

AI or Not?

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Weather records

Earlier observations were used to fit the forecasting model.



2. Forecast

The learned model outputs 70% rain, not a guaranteed shower.



3. Display rule

70 is at least 60, so the written rule displays TAKE ONE.



4. Final decision

The teacher checks current conditions before distributing umbrellas.

CHANGE ONE THING

Keep the forecast but raise the display boundary to 80%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Weather records	Earlier observations were used to fit the forecasting model.
2. Forecast	The learned model outputs 70% rain, not a guaranteed shower.
3. Display rule	70 is at least 60, so the written rule displays TAKE ONE.
4. Final decision	The teacher checks current conditions before distributing umbrellas.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

The **fitting** record supports calling the forecasting stage learned. Comparing 70 with 60 is an explicit rule. The final action belongs to a person.

Merely seeing TAKE ONE would not reveal any of these **mechanism**s; the **process** notes supply that **evidence**.

What this example does not establish

A correct rule comparison does not establish that rain will occur or that **70%** forecasts are followed by rain on about 70 out of 100 comparable days.

Change one thing in the pictured case

Change: Keep the forecast but raise the display boundary to **80%**.

Prediction: The display no longer says TAKE ONE.

Reason: 70 is below 80; changing a display rule does not refit the model.

Ask while showing the picture

Reveal the picture one step at a time

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Use the board sequence and dialogue to model reasoning, reduce help, and check every learner before discussion.

Board sequence

1. Put the facts on the board

Write the given facts: Six devices are a motion-triggered lamp, a trained photo sorter, a remote-controlled car, a calculator, an unspecified smart speaker and an unspecified robot vacuum.

2. State the decision to explain

Write the question to explain: Classify all six using **mechanism evidence**. Which two need **more information**?

3. Reveal the worked reasoning

Reveal after **prediction**s: The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details.

Products can combine **mechanism**s. A robot body or smart label proves none of them.

4. Hold back the independent task

Keep for an independent attempt: A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**?

What **evidence** supports each classification?

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Prediction stage is learned because labelled examples were used to fit the model; album placement follows the stated **written rule**; the owner move is **human control**. Inspect the predicted label and the album action record.

A correct answer from a speaking device does not identify the **mechanism** because recordings, rules, **learned model**s and human messages can all produce speech.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 8 minutes	Ask the clock prerequisite question. Record observed and inferred statements in separate columns. Read two examples aloud without praising a guess as a fact.
8 to 23 minutes	Read the bus case. Trace each row of the diagram with a finger. Ask which decision each row makes. Say that a product can combine mechanisms.
23 to 35 minutes	Model the red-display question using claim, relevant evidence and conclusion. Cover the final table row and ask students to justify it.
35 to 45 minutes	Pairs complete the photo-library explanation and W1. Listen for a distinction between a predicted label and an album action.
Session 2 start 0 to 8 minutes	Retrieve the distinction with the speaking-device exit question. Revisit the diagram if students rely on appearance.
8 to 23 minutes	Complete W2 and W3 in pairs. Each partner must offer a different possible cause of a misplaced photo or direction change.
23 to 38 minutes	Collect W4 as an independent assessment. Do not display its solution or substitute a hinted answer for an independent attempt.
38 to 45 minutes	Discuss one anonymised explanation and set W5 as extension or the next opening task. Record which prerequisite needs reteaching.

Reduce help deliberately

Completion practice

Completion task: Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

Reasoning to listen for

Teacher check: **Prediction** is learned because it was fitted using labelled pictures. Album placement is a **written rule**; the owner move is **human control**. Inspect the stored **prediction** and the placement or move record.

A wrong album alone cannot distinguish **prediction** error from a later action.

> The case-specific table above is a focused two-session sequence (90 minutes). Allow additional sessions for the rest of the weekly work; the allocations are planning examples, not validated durations.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Explain that AI also includes symbolic reasoning. In the specific bus

case, the simple colour threshold is automation; the full system contains a fitted predictor.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Point to the learned stage. Does the changed display prove the weather forecast changed?

Reasoned teacher answer

The forecast stage is learned; it remains **70%**. Only the boundary and resulting message changed.

Changed-case reasoning: The display no longer says TAKE ONE. 70 is below 80; changing a display rule does not refit the model.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

Run the evidence investigation, capture actual observations and choose support from the first reasoning error.

Predict first, test honestly, and explain what the **evidence** means.

1. Predict

State your **prediction** before the result is known.

2. Investigate

Complete the task: Sort six devices by **mechanism** and defend two borderline cases.

3. Compare cases

Keep a normal case and an edge case clearly separated.

4. Record honestly

Record what happened—even when it does not match your **prediction**.

5. Explain and plan the next test

Explain the result, one limitation, and the next useful test.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim

Record	What to capture
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

> **Find the unsupported claim**
>
> CLAIM TO REPAIR Talking proves that a machine learned.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

All rules are outside AI: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

A correct **output** proves a **learned model**: Ask whether a calculator, recording or person could produce the same correct **output**. Require **evidence** about **fitting** or reasoning.

Unknown is a bad answer: Accept unknown when the learner identifies a specific missing fact and a useful way to obtain it.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the keys after the relevant attempts. Previously discussed cases are practice; the reserved independent transfer question is the greenhouse case.

Worked case answer

Teacher key · Worked case answer

The lamp uses a fixed rule; the photo sorter uses a **learned model**; the car follows human commands; the calculator performs arithmetic. The speaker and vacuum need **mechanism** details. Products can combine **mechanism**s.

A robot body or smart label proves none of them.

Independent question

Independent question · withhold the key

A toy follows your steering buttons. A photo app learned from labelled pictures. A talking clock follows a written timetable. Which uses a **learned model**? What **evidence** supports each classification?

Expected independent answer

Teacher key · Expected independent answer

The photo app uses a **learned model** because **training** examples shaped its behaviour. The toy is human-controlled; the clock follows fixed instructions. Speech and movement alone do not establish AI.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Talking proves that a machine learned.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

The **prediction** uses a **learned model**; the colour uses a written threshold; the override is **human control**. The red display alone reveals no **training evidence**. Ask how the delay estimate was produced.

One product can combine **mechanism**s; simple automation does not define all of AI.

Explanations for the additional workbook examples

W1 Separate observations from explanations

Workbook question · W1

A door opens when a visitor approaches. Its label says intelligent entrance. List two **observation**s and write one **mechanism** question whose answer would help classify the opening decision.

Teacher reasoning · W1

Observations include that the door opened when the visitor approached and that its label says intelligent entrance.

A useful question asks whether a written sensor threshold, a model fitted on examples or a person controls opening. The label is an observed marketing claim, not **mechanism evidence**.

W2 Complete the photo library explanation

Workbook question · W2

Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album. Name the first two records you would inspect.

Teacher reasoning · W2

Prediction is learned because it was fitted using labelled pictures. Album placement is a **written rule**; the owner move is **human control**. Inspect the stored **prediction** and the placement or move record.

A wrong album alone cannot distinguish **prediction** error from a later action.

W3 Challenge an unsupported conclusion

Workbook question · W3

Maya says a vacuum must learn because it changes direction after hitting a chair. Give two different **mechanism**s consistent with that **observation**. What **evidence** would distinguish them?

Teacher reasoning · W3

A written contact-sensor rule could turn the vacuum, or a learned navigation model could choose a turn. A person could also control it.

Documentation of the route-selection **process** or a controlled trace could distinguish these **mechanism**s. Direction change alone is insufficient.

W4 Investigate a new hybrid system

Workbook question · W4

A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

Teacher reasoning · W4

The forecast is learned-model **prediction**; the message is a written threshold; vent opening is the caretaker's decision. The message may correctly follow a forecast that is itself inaccurate.

Compare the forecast with a predefined temperature **observation** at the relevant place and time before judging **prediction** accuracy.

W5 Design an evidence card

Workbook question · W5

Invent one product containing two **mechanism**s. State the supplied facts, draw an information path and write one claim the facts support and one they do not. Do not use real classmates' personal information.

Teacher reasoning · W5

Accept a coherent case such as a trained speech recogniser supplying text to a written keyword rule. The support must connect to supplied facts, and the unsupported claim must exceed them, such as assuming perfect recognition.

Check that the learner distinguishes the two decisions rather than merely naming two devices.

Assess independent transfer

Selected case: A school greenhouse estimates tomorrow's temperature using a model fitted on historical weather. A **written rule** displays CHECK PLANTS above 30 degrees C. A caretaker chooses whether to open a vent.

Identify each **mechanism** and explain why a correct CHECK PLANTS message would not prove an accurate temperature **prediction**.

Selected case answer

Teacher key · Selected case answer

The forecast is learned-model **prediction**; the message is a written threshold; vent opening is the caretaker's decision. The message may correctly follow a forecast that is itself inaccurate.

Compare the forecast with a predefined temperature **observation** at the relevant place and time before judging **prediction** accuracy.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

> Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect the explanation to the project, plan extensions and revisit prerequisites where needed.

Helpful Classroom Sorter

Choose whether the Helpful Classroom Sorter needs a person, fixed rule, or **learned model**; justify the smallest safe solution.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 2: **Input**, **Process**, **Output**. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Read the cases aloud and let students use arrows or spoken explanations. Supply the three **mechanism** labels if vocabulary is the barrier, but record that support.

Extension students should design two **mechanism**s producing the same **output** and propose **evidence** that would distinguish them.

For W4, require all three components classified with their supplied **evidence** and an explanation that a downstream rule can operate correctly on a wrong estimate. A label without supporting **evidence** is developing.

If the learner says the display proves accuracy, return to the two bus rows and ask what each row actually checks. After feedback, assess a fresh case rather than awarding independent mastery for the corrected W4 answer.

Week 2 follows how information is represented and changed between components. Keep the three-row bus diagram: the delay number and the warning colour are different representations of related information.

Plan the delayed return

Suggested return points in this level: ai-1/2, ai-1/4, ai-1/7. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For the Helpful Classroom Sorter, annotate which decisions follow written instructions and which, if any, were fitted from examples.

Prepare, teach and assess

> Use this preparation summary alongside the detailed sections; it is a checklist, not a replacement lesson.

Teaching target

Teaching target: Distinguish an observed result from an explanation of how it was produced

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session	Plan	Resources
Session 1 · Understand · 45 min	0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.	Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.
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Before class, solve the supported practice: Use the photo-library case from the lesson. Classify the **prediction**, album-placement rule and owner move. A bicycle photo is in the wrong album.

Name the first two records you would inspect.

Supported practice answer

Reasoned practice answer: **Prediction** is learned because it was fitted using labelled pictures. Album placement is a **written rule**; the owner move is **human control**. Inspect the stored **prediction** and the placement or move record.

A wrong album alone cannot distinguish **prediction** error from a later action.

Check every learner before discussion: Use the worked case for **Evidence** and decision **mechanism**s. Proposed explanation: “All rules are outside AI”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Explain that AI also includes symbolic reasoning. In the specific bus case, the simple colour threshold is automation; the full system contains a fitted predictor.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Keep the forecast but raise the display boundary to **80%**. Require a **prediction** and a reason before comparing. Expected result: The display no longer says TAKE ONE.

Reason: 70 is below 80; changing a display rule does not refit the model.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, review links and portfolios purposefully. Coding guidance applies when a later lesson contains a lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For lessons with executable code

> Week 1 has no Python lab. Retain this routine for later coding weeks.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

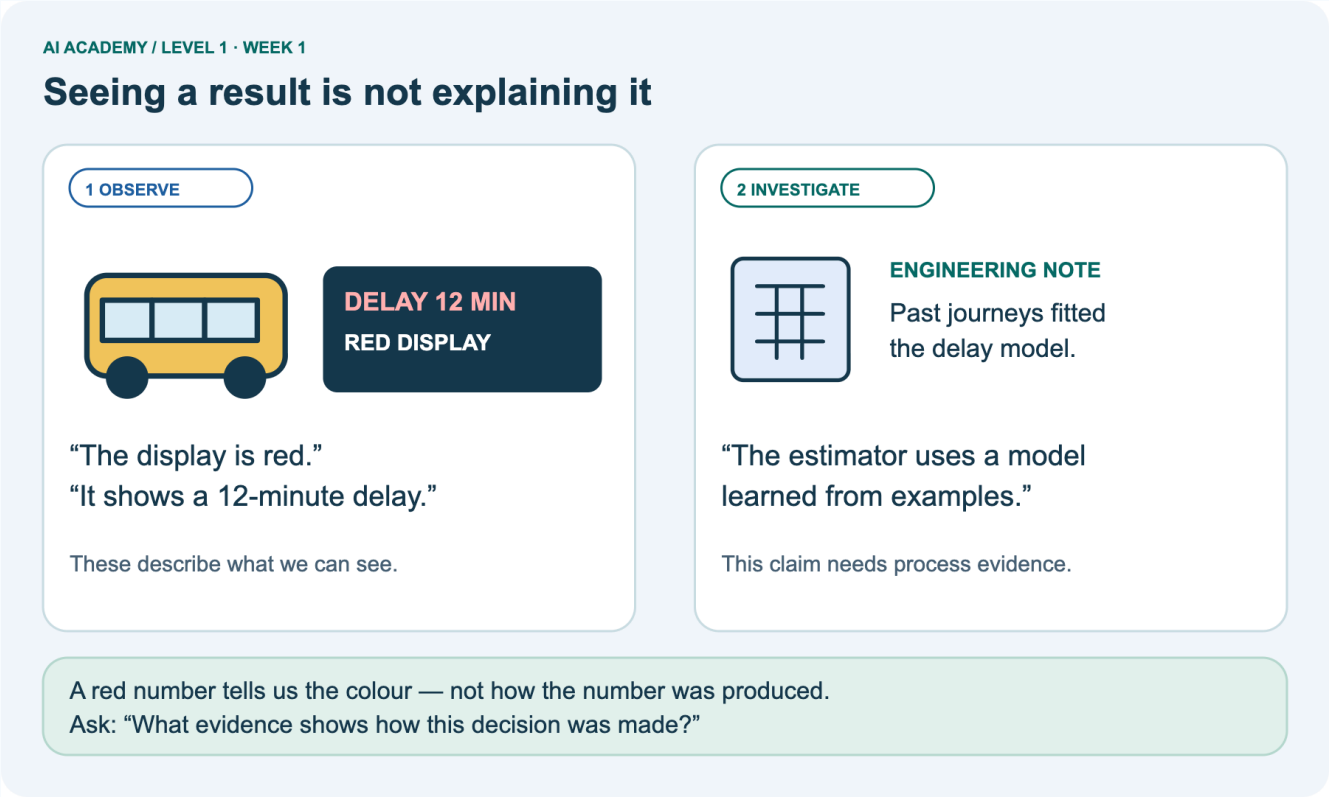
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one graphic at a time, ask for a prediction, then use the labelled trace and teacher reasoning to discuss it.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Seeing a result is not explaining it



Follow the labelled picture with the matching explanation. Open the vector diagram

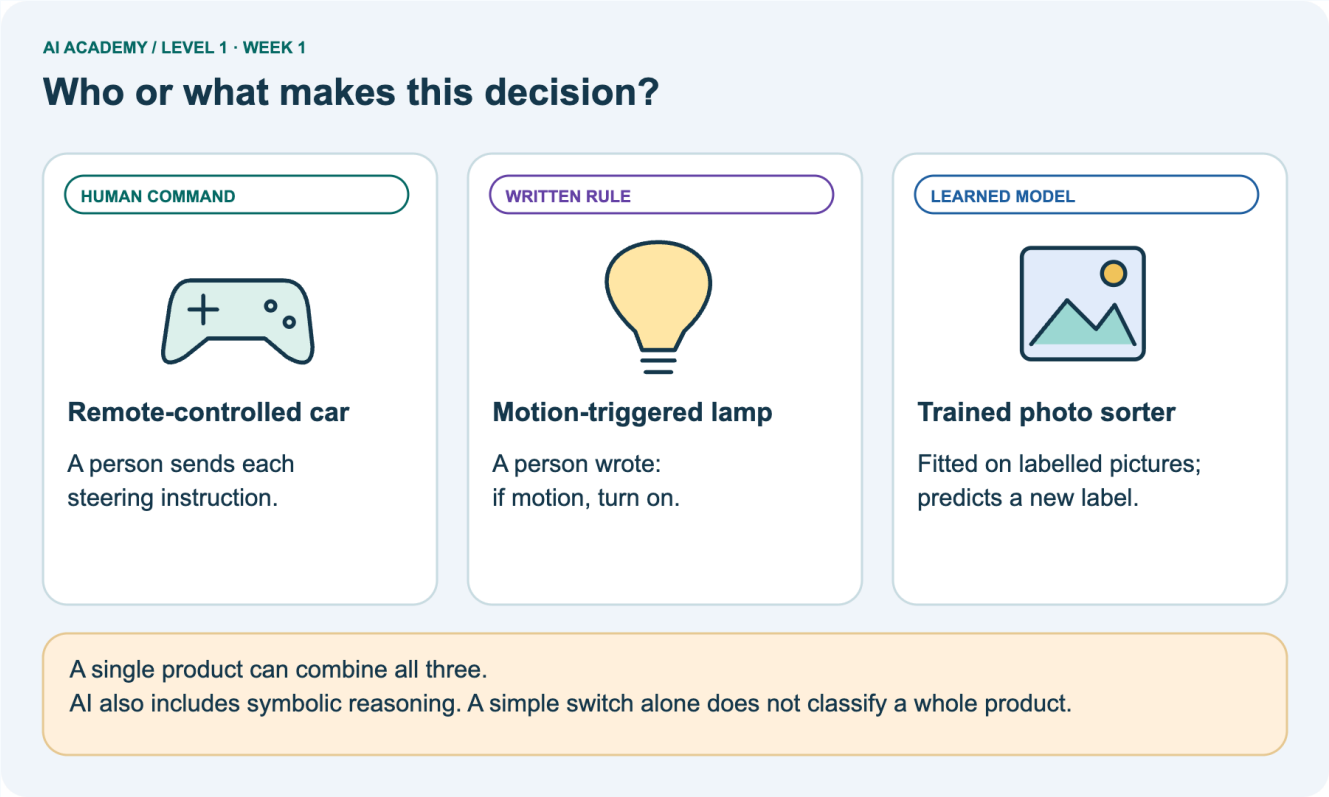
A bus has a red display showing a 12-minute delay. That is an **observation**. A supplied engineering note says past journeys fitted the delay model; that is **evidence** about its **mechanism**.

> Name one **observation** and one claim that needs more **evidence**.

Discuss while showing the picture

Teacher prompt: Name one **observation** and one claim that needs more **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Who or what makes this decision?



Follow the labelled picture with the matching explanation. Open the vector diagram

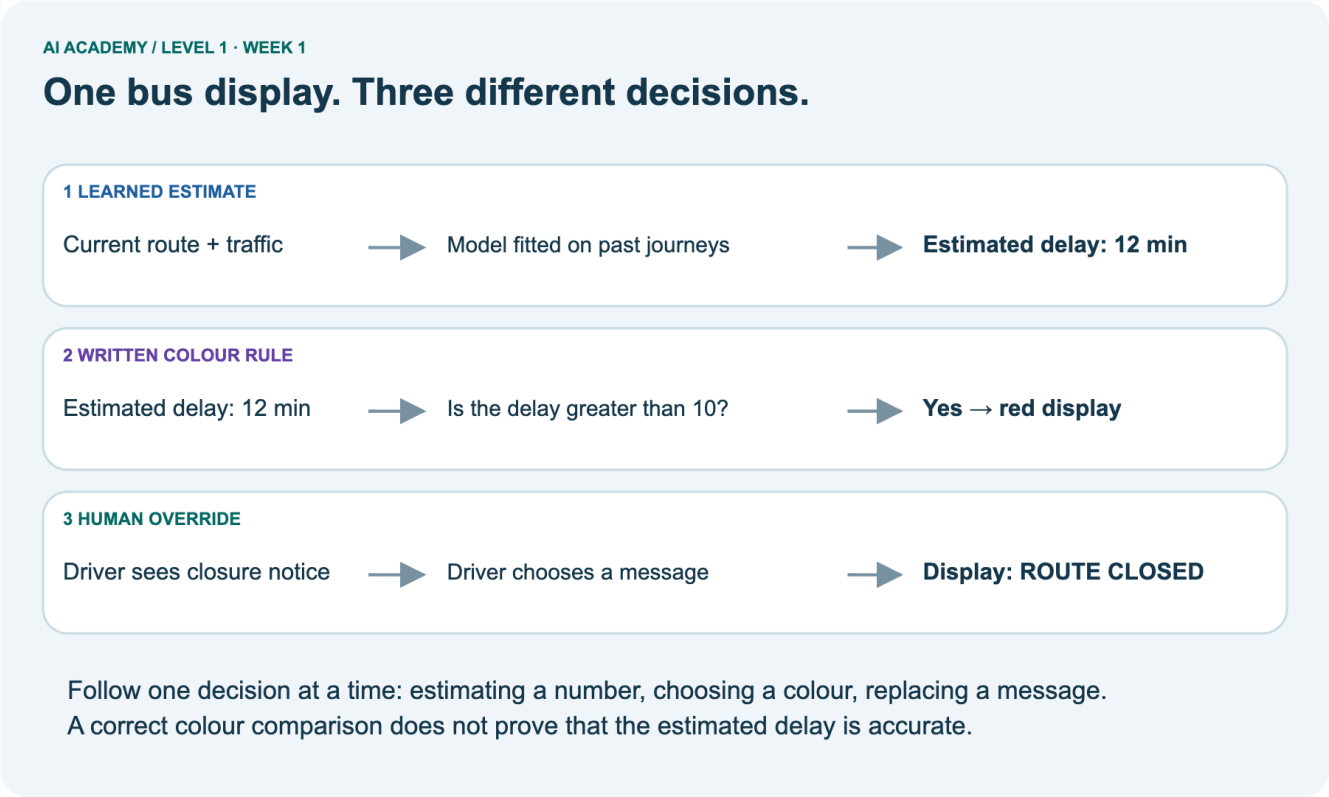
A remote-controlled car follows human commands, a motion lamp follows a **written rule**, and a trained photo sorter applies a **learned model**. A product can combine **mechanism**s. AI also includes symbolic reasoning.

> Point to who or what makes each decision.

Discuss while showing the picture

Teacher prompt: Point to who or what makes each decision. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One bus display. Three different decisions.



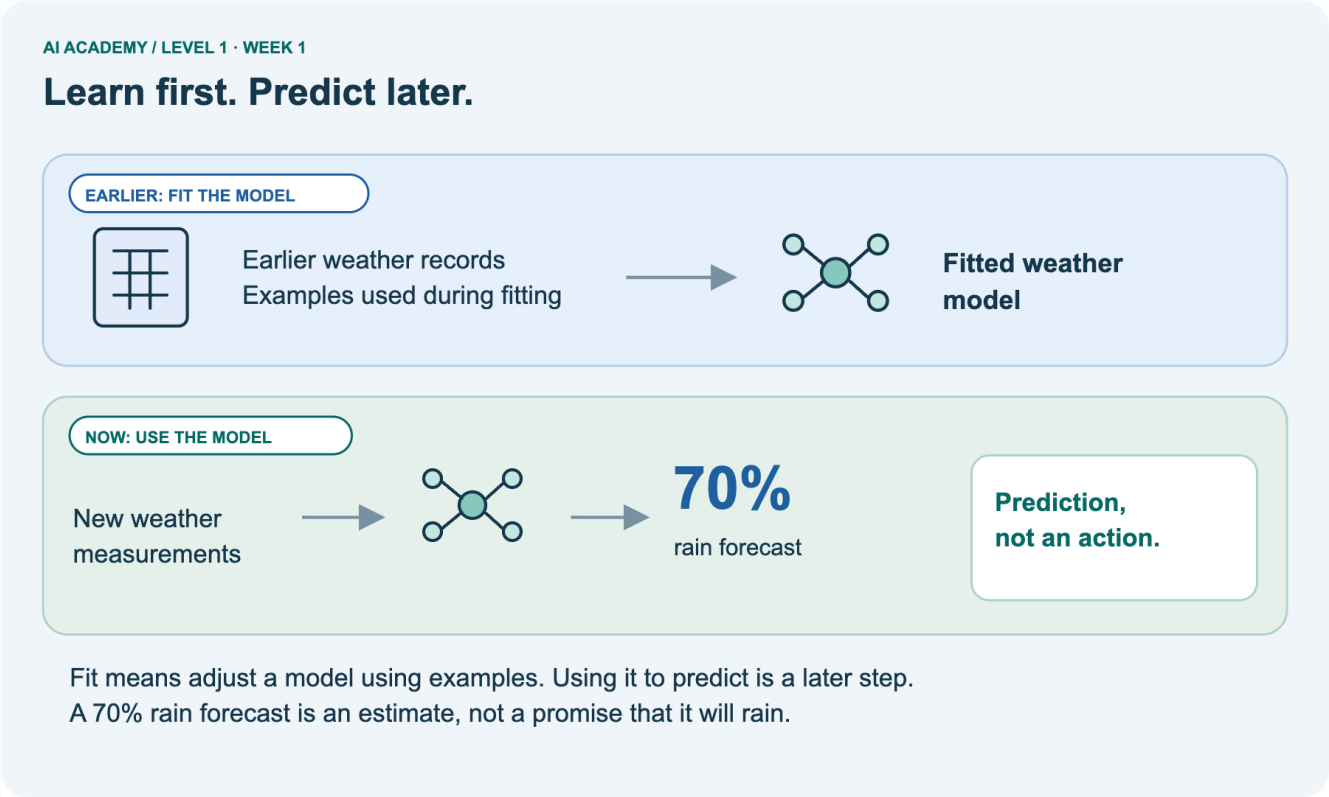
Follow the labelled picture with the matching explanation. Open the vector diagram

Three separate rows trace the bus estimate, display colour and driver override. The fitted model estimates **12 minutes**; the written greater-than-10 rule makes the display red; the driver can choose ROUTE CLOSED.

> Trace the number, colour and override separately.

> **Discuss while showing the picture**
>
> Teacher prompt: Trace the number, colour and override separately. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Learn first. Predict later.



Follow the labelled picture with the matching explanation. Open the vector diagram

Earlier weather records are used to fit a model. Later, new measurements are passed to that fitted model to obtain a 70 percent rain forecast. It is an estimate, not a guarantee.

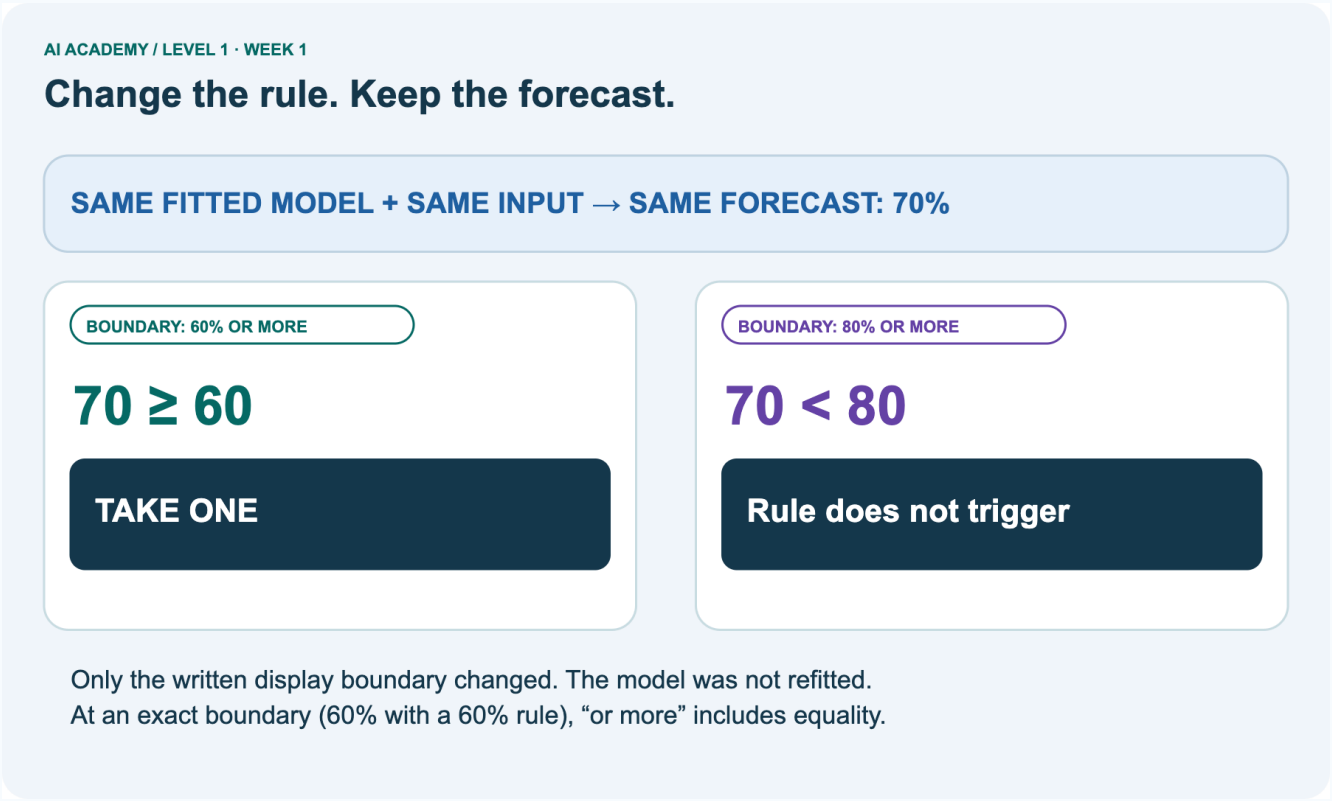
> Which steps use earlier examples, and which use a fitted model?

Discuss while showing the picture

Teacher prompt: Which steps use earlier examples, and which use a fitted model? Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Changed-case reasoning: 70 is below 80; changing a display rule does not refit the model.

05. Change the rule. Keep the forecast.



Follow the labelled picture with the matching explanation. Open the vector diagram

With a 70 percent rain forecast, a 60 percent-or-more rule displays TAKE ONE. An 80 percent-or-more rule does not trigger that message. Only the boundary changed; the fitted model and forecast did not.

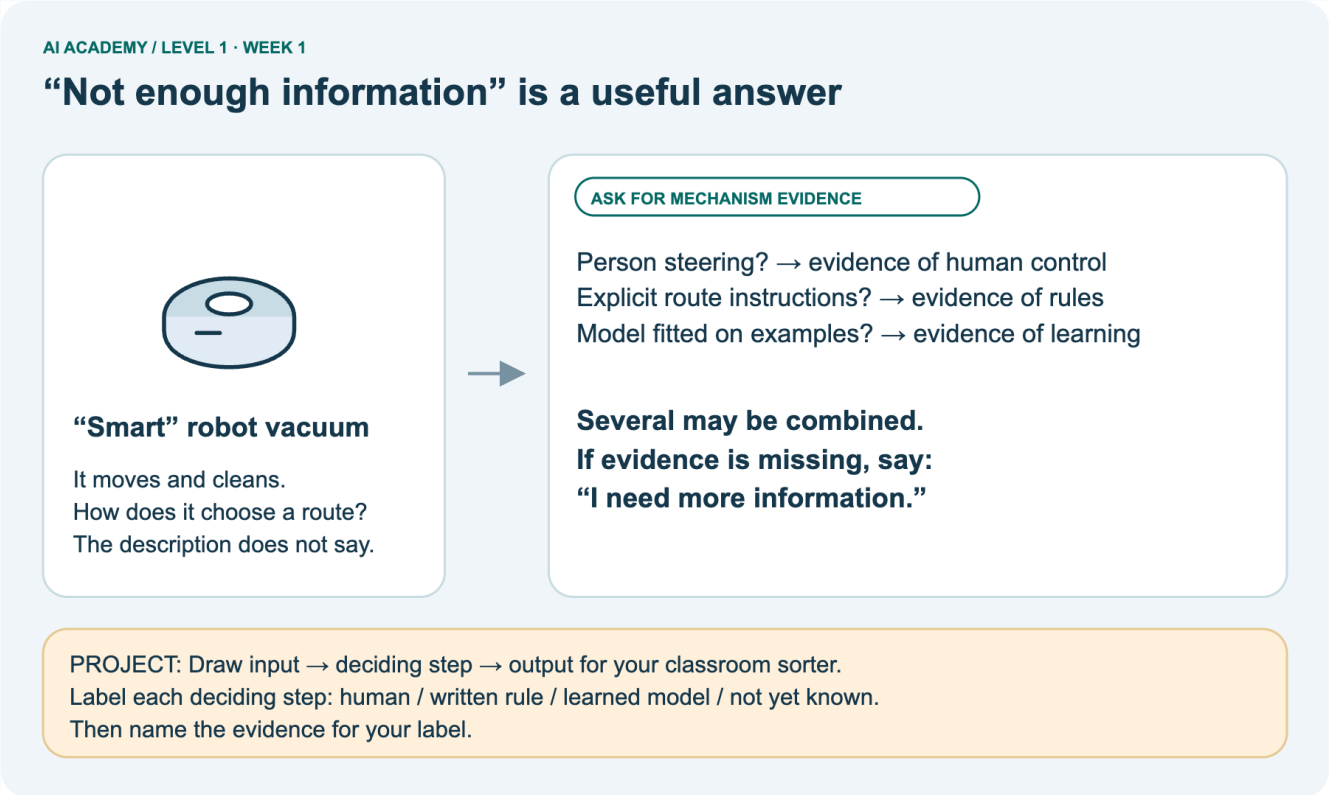
Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**?

Discuss while showing the picture

Teacher prompt: Keep the forecast fixed at **70%**. What changes when the boundary becomes **80%**? Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

> Keep the stated scope: A correct rule comparison does not establish that rain will occur or that **70%** forecasts are followed by rain on about 70 out of 100 comparable days.

06. “Not enough information” is a useful answer



Follow the labelled picture with the matching explanation. Open the vector diagram

An unspecified smart robot vacuum moves and cleans, but its route **mechanism** is unknown. Ask for **evidence** of human commands, **written rule**s or model **fitting**, and allow combinations. Apply this reasoning to a classroom sorter.

> Which **evidence** would change your answer about the vacuum?

Discuss while showing the picture

Teacher prompt: Which **evidence** would change your answer about the vacuum? Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The forecast stage is learned; it remains **70%**. Only the boundary and resulting message changed.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.